

NEWSLETTER

MAY 2026

THE PATH OF EM-06

FORMULA STUDENT
SPAIN

FORMULA STUDENT
EAST

TEAM UPDATE

SALÓ DE
L'ENSENYAMENT

INTERVIEW WITH
THE TEAM LEAD

EUSS MOTORSPORT

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INTRODUCTION

THE PATH OF EM-06

We are pleased to share with you the current status of a project that has been built, hour after hour, in the EUSS workshop. The EM-06 is no longer just a set of calculations and designs; it is the tangible result of the effort of a team of students who seek to get the most out of every resource and every lesson learned.

These months have been full of hard work and realistic decisions. We have learned that the best engineering is not always the most complex, but the one that is executed with the most care and coordination. From reducing weight in the chassis to optimizing the electronics, every small improvement is a collective victory that prepares us for the challenges that await us this summer in **Montmeló and Hungary**.

We are fully aware that this car would not touch the asphalt without your trust. Every component you give us, every technical advice and every show of support are the pieces that really make EUSS Motorsport move forward. For us, your support is the guarantee that the effort in the workshop has a meaning that goes beyond the competition.

Thank you for being part of this learning and for making our work a reality.

The EUSS Motorsport team



TEAM UPDATE

The evolution of the EM-06 is the result of a collective effort where learning and humility have been our best tools. In this section we detail the progress of each department: from reducing the weight of the chassis to improving electronic reliability.

We are a small team that learns with every step, and we are fully aware that none of these technical achievements would be possible without your trust and support. Each improvement that we detail below is also a success shared with you.

This is how our project is growing:

TOTAL CAD DEPARTMENT UPDATE

• Description of the work performed:

The full CAD design has been completed from scratch, integrating all major parts and closing the vehicle packaging. The main focus has been technical coordination between departments to ensure that critical parts, such as those for Vehicle Dynamics, are integrated correctly in order to advance to the welding phase.

• Key achievements and metrics:

- Weight reduction: The new chassis model weighs 10 kg less than the previous year's.
- Structural efficiency: The same resistance and torsional rigidity as in the previous version have been maintained, but with a lighter structure.
- Material optimization: A selection of tubes has been made looking for the maximum inertia ratio with the lowest possible weight.
- Center of gravity: The packaging design has met the target position defined by the dynamics department.

• Featured sponsors:

- Motor Bike's contribution: Thanks to Motor Bike's financial support, the team can afford the parts and processes necessary to manufacture with the greatest possible precision.
- Elimination of human error: This support is essential to outsource processes that eliminate the team's margin of manual error, guaranteeing superior technical execution to deliver the car with guarantees.
- Investment in development: Although in previous phases progress was made with own resources, the entry of these funds allows the quality of the final components to be scaled up.

• Next steps:

- Technical Coordination: The management role has been consolidated to synchronize all technical departments and ensure that the parts arrive on time.
- Start of manufacturing: Once the design and CAD are closed, the manufacturing phase of the chassis and the rest of the car's elements has begun.
- Physical Integration: The next big challenge is the assembly of the Powertrain and engine components within the chassis according to the stipulated design.



TEAM UPDATE

POWERTRAIN DEPARTMENT UPDATE

- Description of the work performed:

The design from scratch of the new accumulator (power battery) has been completed. This component is fundamental to the vehicle's performance and represents the second iteration of the battery developed by the team. In addition, structural improvements have been made to the steering and braking of the cart to increase its operational safety.

- Key achievements and metrics:

- Power increase: The battery has increased its total power by 10 kW thanks to the new design and cells.
- Capacity: A capacity of 8 kWh has been achieved.
- Dimensional optimization: The width of the battery has been reduced to 423 mm, eliminating one of the main limitations for the chassis design.
- Weight efficiency: These improvements have been achieved while maintaining the total weight of the component.

- Featured sponsors:

- Igus and Miguelez: Fundamental collaborators in the provision of technical wiring.
- Batemo: Key partner for carrying out electrothermal simulation studies of the cells

- Next steps:

- Regenerative Braking: Studies are being carried out to implement this system in the car, which is currently in the validation phase.
- Advanced Simulation: Use of cell models to carry out detailed electrothermal studies.

OPERATIONS DEPARTMENT UPDATE

- Description of the work performed:

The entrance exams for the competitions were held at the beginning of February. These exams are online and the whole team answers them together, usually of high complexity. We have also started working on the static tests of the competitions since Operations this year is in charge of the Designs, Cost and Manufacturing Event and Business Plan Presentation.

- Key achievements and metrics:

- We have entered the competitions we were aiming for: FS Spain and FS East.

- Featured sponsors:

- Meridiana Rent a Car: Discount for the truck that will go to Hungary to carry all the team's equipment.

- Next steps:

- Quizzes: We close with all the objectives fulfilled regarding the exams. Number of simulations, level acquired and results.
- Design: The designs try to defend the aspects of the car presented to the competition judges. They begin to prepare and collect information.
- Cost: It tries to create a list with all the components and costs of the car. It requires great management of the team's resources.
- BBP: It tries to present a business plan that includes some aspect of the car. We are currently studying previous documentation from other years to improve.

TEAM UPDATE

ELECTRICAL SYSTEM DEPARTMENT UPDATE

- Description of the work performed:

We have implemented CANBUS throughout the car to improve data acquisition and diagnosis. All the plates have been renewed including component improvements, system simplification and increased security. Giving priority to system reliability.

A custom program has been created to be able to monitor in real time the status of all the car's parameters, providing great ease when diagnosing problems that may arise. At the same time, a proprietary datalogger has also been created to be able to collect all the car's data and analyze it later.

An improvement has been made to the vehicle's wiring, reducing and simplifying it in order to minimize the problems that may arise and making it easier to assemble and handle.

- Key achievements and metrics:

- Reduced wiring: The new harness reduces the number of cables present by approximately 40%, reducing the possibility of connection problems.
- Reduced dimensions: The new PCBs have smaller dimensions in order to make the boxes and receptacles smaller and easier to locate in the car.
- Total communication: The implementation of CANBUS allows complete acquisition of all vehicle data in real time, creating the possibility of being able to analyze them later and make improvements more quickly.

- Featured sponsors:

- Wattius: Continues to support us in the vehicle's BMS, ensuring good management of the high-voltage battery
- Würth Elektronik: Provides us with electronic components for our PCBs, with good product quality and guaranteed good operation.
- Pure PCB: Another year brings us the quality of the manufacturing of all the PCBs of our car.
- LEMO: The signal connectors of a car are extremely delicate, due to exposure to high vibrations and bad conditions, which is why it is necessary to have quality connectors like those from LEMO.

- Next steps:

- Change of microcontrollers of all switchboards with a more professional product and present in the industry.
- Improvement in the transfer of knowledge and documentation of the processes and decisions of the department.
- We want to continue working with the clear objective of reliability and rapid diagnosis to guarantee that the possibility of error is always minimal



TEAM UPDATE

VEHICLE DYNAMICS DEPARTMENT UPDATE

- Description of the work performed:

The department has been divided into three major work areas: transmission, dynamics and brakes. In the dynamics area, the design and manufacture of a new wheel assembly has been completed, including knuckles and hubs. As for the brake system, a completely new design has been made from scratch that integrates a new pedal, pumps, calipers and brake discs. Currently, the team is waiting to receive all the parts to begin assembly on the new chassis.

- Key achievements and metrics:

- Weight reduction in Dynamics: With the new design of the spindle and the spark plug, the weight of the wheel assembly has been reduced by 60%.
- Chassis optimization: The new brake system has allowed the length of the nose to be reduced by 6 cm, which means a 10% reduction in the total length of the chassis.
- Braking system efficiency: A weight reduction of 60% has been achieved in the rear brake system and 50% in the front.
- Component innovation: A configuration of two calipers per wheel on the front axle and one per wheel on the rear has been adopted, using much lighter calipers than the previous ones.

- Featured sponsors:

Expansion of Sponsors: This year, we have key collaborations for manufacturing and validation:

- Blume Prot: They have machined a large part of the wheel assembly, including all the hubs and a specific part for the transmission.
- Asorcad: They have supported by performing 3D scans of several parts of the transmission
- ISR Brakes: They have collaborated with a significant discount on the brake calipers.

- Next steps:

- Integration Challenge: The main challenge has been the miniaturization of all the components to fit a complex brake system in a very small space.
- Assembly Phase: The next immediate step, once the knuckles and the rest of the parts are received this week, is to mount everything on the welded chassis to verify that the design fits perfectly.



TEAM UPDATE

AERODYNAMICS DEPARTMENT UPDATE

- Description of the work performed:

The design of the EM-06 aerodynamic package has been completed, significantly improving its performance compared to last year. The engine and inverter cooling has also been designed, thanks to a new team sponsor. During this period, manufacturing tests have been carried out to achieve a better result in the final parts of the car.

- Key achievements and metrics:

- Aerodynamic Efficiency: The new vehicle improves the efficiency of the previous one, going from an efficiency of -0.97, to more than double.
- Manufacturing progress: The Aerodynamics department has taken a step forward by once again using molds designed to be manufactured on CNC machines for better surface finishes.
- Results optimization: A basis for comparing simulation results has been established, exceeding the number of simulations carried out, exceeding the 82 carried out with the EM-05, with 200 different design iterations for the EM-06.
- Department training: With the challenge of significantly expanding the department's staff, 13 people have been reached, who have received different training in the design and manufacture of carbon fiber parts.

- Featured sponsors:

Expansion of Sponsors: This year, we have renewed and added new sponsors:

- Mel Composites: Sponsor that renews for the 5th season with the team, becoming a diamond category. They help us with the manufacture of molds to manufacture the parts and with all types of consumables to manufacture carbon fiber.
- Castro Composites: Consumables, highlighting the vacuum bag.
- Easy Composites: Specific consumables and manufacturing tutorials.
- Boellhoff: Fasteners.
- Pudripedia: Economic Sponsor.
- FYK: Economic Sponsor.
- Starcke: Abrasive material for the manufacture of molds and parts.
- Alphacool: Material used for car cooling.
- GAS: Economic Sponsor.
- Blackfabric: Carbon fiber fabrics.

- Next steps:

- Manufacturing season: With the arrival of the season of competitions and tests, the department's manufacturing season has begun.
- Design Preparation: We have begun to study how the Engineering Design test will be presented in the different competitions in which the team will compete this summer.



TEAM UPDATE

MARKETING DEPARTMENT UPDATE

• Description of the work performed:

A deep renovation of the team's image and digital strategy has been carried out. The website has been completed in three languages (Catalan, Spanish and English) and content creation has been professionalized by moving from Canva to Adobe tools (Illustrator and Photoshop). In the area of social networks, the next 15 Instagram posts have been planned, the white background for product and department photography has been standardized, and the presence on LinkedIn has been improved.

Regarding merchandise, the design and selection of pieces (T-shirts, polo shirts, sweatshirts, jackets and fleeces) has been closed, implementing a new order system with advance payment to optimize stock. Finally, the growth of the team has been managed with the incorporation of new profiles and all sponsorship acquisition documentation has been updated.

• Key achievements and metrics:

- Expansion of Sponsorships: Agreements have been closed with 8 new sponsors, highlighting the alliance with Red Bull (supply of 150 cans per month) and a 30% discount with WürthModyf.
- Merch Optimization: A competitive cost price of €7 per shirt has been achieved and a profit system has been established for the team in sales to the public.
- Digitalization and Coverage: Fully operational website in 3 languages and closed planning of 10-15 publications to maintain consistency on networks.
- Data Management: Creation of a new Sponsors Book from scratch and migration of the entire contact database to an optimized Excel format.

• Featured sponsors:

Sponsorship Expansion: This period has focused on closing logistical and supply partnerships:

- Red Bull: Official beverage partner for the team (initial delivery of 500 cans).
- Würth Modyf: 30% discount on material and immediate order logistics.
- Loctite: Collaboration on digital content and technical product.

• Next steps:

- Merchandise Production: Collection of payments, size testing and shipping of the final order to receive the material before the end of June.
- Vinyl and Technical Design: Finalization of the designs of the technical t-shirt and preparation of the vinyl of the car from the CAD sent this week.
- Institutional Presence: Management of talks and stands at the UAB, Sant Jordi and preparation of the Roll-out.



Polar jacket

Basic white t-shirt



EUSS MOTORSPORT AT THE 2026 SALÓ DE L'ENSENYAMENT

The EUSS Motorsport team traveled to the Montjuïc venue to participate in a new edition of the Saló de l'Ensenyament. They were intense days where our stand became an indispensable meeting point for automotive and engineering enthusiasts.

A VISUAL MAGNET: OUR SINGLE-SEATER

If anything made a difference in the university pavilion, it was the presence of our car. In an environment dominated by brochures and screens, EUSS Motorsport opted for the tangible reality of engineering: we were one of the few teams to bring a real vehicle to the Show.

This choice was not accidental. The single-seater acted as the best ambassador for our university, attracting the attention of curious and experts alike. For many attendees, discovering that a group of students is capable of designing and manufacturing a machine of this caliber was the first step towards becoming interested in our degree.



CONNECTING WITH THE NEW GENERATIONS

Our work at the stand went far beyond the technical exhibition. We focused on guiding each student according to their stage of life:

ESO Explorers: We explained the path to engineering to 4th grade students, detailing the difference between middle and higher level training cycles and direct access via Baccalaureate.

Baccalaureate Decision: 1st and 2nd year students came with more mature questions. We conveyed our educational model to them: a real balance of 50% theory and 50% practice.

Technical profile: We had a constant flow of young people specifically interested in the areas of Automotive and Industrial Organization, profiles that fit perfectly with the culture of our team.



PRESENCE IN EVENTS

EUSS MOTORSPORT AT THE 2026 SALÓ DE L'ENSENYAMENT

"IS THIS A SUBJECT?": THE REALITY OF FORMULA STUDENT

One of the most recurring questions was whether car construction was part of the official academic curriculum. Our answer was clear and direct: EUSS Motorsport is an extracurricular project driven by passion.

We explained to future students and their families that being part of the team is not a question of credits, but of a vocation for the world of motorsport and a desire to get your hands dirty from day one. Unlike other institutions where access to these projects is restricted to the last years, at EUSS we open our doors to students from the 1st year of their degree. This "early quarry" is what allows us to maintain knowledge year after year.

HUMILITY, INGENUITY AND RESULTS

In the most in-depth conversations with parents who were interested in more pragmatic aspects such as the facilities, teaching staff and scholarship system of our charter university, we never hid our reality.

"We are a small university with a modest budget compared to giants like the UPC, but that is precisely what makes us great".

At EUSS Motorsport, success is not measured only in the final trophy of the Formula Student. It is measured in our ability to optimize resources, in collaborative learning by departments and in the constant evolution of each piece we design. As we explained at the Salon: what counts is not just winning, but demonstrating that with little we can do a lot.



FINAL REFLECTION

We returned to our headquarters in Sarrià with our batteries charged. The Saló de l'Ensenyament was not just a trade fair, but a validation of the work we do daily in the workshop. Seeing the sparkle in the eyes of a 15-year-old student in front of the single-seater we created reminds us why we started this project. Thank you to everyone who shared a moment with us! The future of engineering is in good hands



COMPETITIONS

FORMULA STUDENT SPAIN AND EAST



At EUSS Motorsport we are aware that a single-seater is not built with just parts and cables; it is built with trust, time and shared learning. After a year of constant work, many hours in the workshop and overcoming challenges that have tested us as a team, we are very excited to announce our participation in two of the most prestigious Formula Student competitions this summer. For us, these dates are not just a sporting event, but the moment to validate all the enthusiasm we have invested in the project.

FIRST STOP: HOME



**Circuit de Barcelona-Catalunya
(Montmeló)**



1 - 7 AUGUST

Running in Montmeló is, above all, an act of gratitude. It is the opportunity to show the fruit of our work to our families, friends and, especially, to the local sponsors who have opened their doors to us. We come with the excitement of last year's success and we return with the EM-06 with more desire than ever to make the entire EUSS community that has seen us grow proud.

SECOND STOP: THE INTERNATIONAL CHALLENGE



Hongria (Hungaroring)



23- 29 AUGUST

Arriving in Hungary represents the most ambitious step in our history. We are a humble team that goes to one of the most demanding competitions in the world with the utmost desire to learn. We go there to measure ourselves against the best, but above all to show that with effort, optimization of resources and great teamwork, a small university can go very far.



INTERVIEW WITH THE TEAM LEAD

INTERVIEW WITH GUILLEM CASTELLNOU

Within a project as complex as Formula Student, we often look at the engine power or the aerodynamics of the car, but there is a fundamental piece that is not seen on the track: human and logistical management. We speak with Guillem Castellnou, Team Lead and Operations Manager, who with a realistic vision and a lot of dedication, explains what it is like day to day coordinating a team of passionate people to get as far as possible.

• MANAGEMENT AND FACILITATION

Q: How do you ensure that the members of the technical departments have everything they need in the workshop without having to waste time on bureaucracy?

- The key is face-to-face and constant communication: every week we all meet at the university.
- We take advantage of these meetings to discuss the progress of the tasks and update purchases in a general way, preventing people from getting lost in technical details that do not concern them.
- In this way, we try to keep the information flowing and so that the technical department can focus exclusively on the car.

• ACTIVE LISTENING

Q: When a department head asks you for something impossible or last-minute for the car, how do you handle it with humility so as not to generate conflicts?

- I understand that the Team Lead figure is to organize, but I don't know more than the specialists in each department.
- If a last-minute request arises, we check two variables: if it is within the budget and if it fits into the calendar (Gantt).
- I don't impose my word; if an idea seems like it won't arrive on time or is out of budget, we do a small joint intervention with all the coordinators.



INTERVIEW WITH THE TEAM LEAD

INTERVIEW WITH GUILLEM CASTELLNOU

• WELLBEING AND WORK ENVIRONMENT

Q: The manufacturing season is tough. What are you doing from Operations to ensure that the atmosphere in the workshop is good and people don't burn out?

- Operations acts as logistical support: we prepare the material, maintain order and define clear schedules to be more efficient.
- Experience tells us that a mechanic works much better if he has someone to save him time by looking for parts or managing nomenclatures.
- We take care of these administrative tasks so that the technical departments don't have to wear themselves out with them.

• INTERNATIONAL VISION AND CHALLENGES

Q: Going to Hungary is a huge logistical challenge. What is your biggest fear about the trip and how are you preparing yourself in case something goes wrong?

- It's the biggest challenge in our history: 2,000 kilometers one way across several countries to the east.
- For safety, we have reduced the expedition to 20 people and are planning everything in detail: stops, gas stations depending on the car and fuel prices by country.
- We have even located the closest airports along the route to have a "plan B" in case a vehicle gets stranded or someone needs to return urgently.





INTERVIEW WITH THE TEAM LEAD

INTERVIEW WITH GUILLEM CASTELLNOU

• SAFETY AND ORDER IN THE WORKSHOP

Q: How are you ensuring that the EUSS workshop is a clean and safe place where everyone respects the rules without having to be the "policeman" of the team?

- It is not a matter of imposition, but of collective awareness and respect for the university and the professors with whom we share the space.
- It has been shown that there is much more production when the workshop is tidy and the floor is clean.
- We try to make the space available and practical for everyone, avoiding the accumulation of components on the work tables.

• TALENT MANAGEMENT AND LEARNING

Q: Operations has many small but vital tasks. How are you teaching new members to learn how to manage suppliers or invoices from scratch?

- We work on a "generational change" project where we already have a clear picture of who the next leaders will be.
- We do a very organic transfer of knowledge from mid-season, so that when the time comes to be responsible they already have months of experience.
- New members learn the Formula Student rules from scratch, even if it is difficult for them to understand the whole team structure at first.

Q: What do you think is the most important thing you have learned this year as an Operations Lead that you would tell your past self before starting?

- I would tell him to enjoy his last year more and not to burn himself out so much.
- I have learned that the Team Lead is a necessary figure of responsibility, but that success depends on the autonomy and specialization of each coordinator.
- It is vital to be present in the workshop; productivity and contact with the team are lost if you try to work only from home.



YOU ARE THE ENGINE THAT DRIVES US

We have reached the end of this newsletter, but the work in the workshop does not stop. As you read these lines, some member of the team is adjusting a cable, polishing a mold or reviewing a simulation to ensure that the EM-06 lives up to the trust you have placed in us.

At EUSS Motorsport we know that the result on the track is important, but what really defines us is the journey: the hours of learning, the resilience in the face of unforeseen events and the network of collaborators who make this project viable year after year.

Do you want to experience the process with us? We encourage you to follow our day-to-day life more closely. Where the technical data in this newsletter does not reach, you will find the reality of our workshop, our successes and, also, our learnings:

Thank you for sharing this passion with us. See you on the next bend in the road, ever closer to the starting line in Montmeló and Hungary.

The EUSS Motorsport team.





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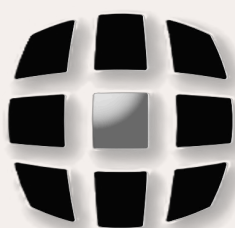


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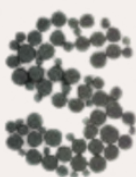
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